Anxin (Bob) Guo

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Education

Northwestern University, Evanston, Illinois September 2023 – June 2028 (expected)
Ph.D. student in Computer Science. Cumulative GPA: 3.867/4.000.
Northwestern University, Evanston, Illinois September 2019 – June 2023
B.A. in Mathematics and M.S. in Computer Science. Undergraduate cumulative GPA: 3.821/4.000.

Research Interests

Theoretical computer science and the foundations of machine learning, with interests in the capabilities and limitations of learning and computation, and how language models learn, memorize, and reason.

Research Projects and Papers

Publications

Hallucination is a Consequence of Space-Optimality: A Rate-Distortion Theorem for Membership Testing
Anxin Guo and Jingwei Li.
International Conference on Machine Learning (ICML), 2026, spotlight, top 2.2%. arXiv.

Agnostic Learning of Arbitrary ReLU Activation under Gaussian Marginals
Anxin Guo and Aravindan Vijayaraghavan.
Conference on Learning Theory (COLT), 2025. arXiv; conference version.

To Store or Not to Store: A Graph-Theoretic Approach for Dataset Versioning
Anxin Guo, Jingwei Li, Pattara Sukprasert, Samir Khuller, Amol Deshpande, and Koyel Mukherjee.
IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2024. arXiv; conference version.

Manuscripts

Approximate Syzygies of Smoothed Polynomials and Applications
Aditya Bhaskara, **Anxin Guo**, Tanmay Sinha, Vaidehi Srinivas, and Aravindan Vijayaraghavan.
Manuscript. PDF.

Learning Rules, Memorizing Facts: An Information-Theoretic View of Hallucination
Gengze Xu, **Anxin Guo**, Yifei Hu, Xiangyu Wu, Yong Liu, and Ziqiao Wang.
Preprint, 2026. arXiv.

How Much Do You Trust A Dog? Quantifying Hint-Following in Reasoning Language Models
Anxin Guo, Chirag Pabbaraju, Sara Ahmadian, Kamesh Munagala, and Aravindan Vijayaraghavan.
Preprint, 2026. arXiv.

Regularized Transport Weights: a Principled Approach for Active Domain Adaptation
Anxin Guo, Tanmay Sinha, Dionysis Arvanitakis, Vaidehi Srinivas, Alex Drlica-Wagner, Alexander Ji, and Aravindan Vijayaraghavan.
Manuscript.

Leanify: Verified Proof Compression and Reusable Lemma Discovery
Ruize Xu, **Anxin Guo**, Yi Gu, Tianchong Jiang, Chenxiao Yang, Tianhao Wang, and Zhiyuan Li.
Manuscript.

Copying Under Repetition: From Prefix Matching to Position Hashing
Anxin Guo, Tianlong Huang, Chenxiao Yang, and Zhiyuan Li.
Preprint, 2026. arXiv.

Selected Talks

Simons Institute — Student spotlight October 7, 2026
Hallucination is a Consequence of Space-Optimality.
Workshop: *Trustworthy AI: From Hallucinations to Reliable Autonomy.*

SkAI Works-In-Progress — Joint presentation September 30, 2026
Intelligent Scheduling for Astronomical Surveys.
With Rachel Hur and Tanmay Sinha.

Google AI Study Group — Invited talk September 14, 2026
Why LLM Hallucinate? A Perspective from Space-Optimality.

IDEAL Annual Meeting, TTIC — Lightning talk September 10, 2026
Hallucination is a Consequence of Space-Optimality.

IDEAL Annual Meeting, UIC — Student talk June 9, 2025
Agnostic Learning of Arbitrary ReLU Neuron under Gaussian Marginals.

Northwestern University — CS Theory Seminar February 6, 2025
Agnostic Learning of Arbitrary ReLU Neuron under Gaussian Marginals.

NSF TRIPODS Workshop, TTIC — Lightning talk December 7, 2024
Agnostic Learning of Arbitrary ReLU Neuron under Gaussian Marginals.

Awards and Honors

Ph.D. Student Research Award, Computer Science Department, Northwestern University 2024 – 2025
Barris Award for Outstanding TA, CS 335: Introduction to the Theory of Computing Fall 2024
Junior Career Award in Mathematics, Mathematics Department, Northwestern University 2021 – 2022

Research and Teaching

IDEAL-funded summer research exchange, University of Chicago Summer 2026
Hosted by Chao Gao.
Research Intern / IDEAL-funded summer research exchange, TTIC Summer 2025
Toyota Technological Institute at Chicago. Hosted by Zhiyuan Li.
Teaching Assistant, CS 336: Design and Analysis of Algorithms Spring 2026
Teaching Assistant, CS 335: Introduction to the Theory of Computing Fall 2024

- Gave a guest lecture on busy beaver numbers in addition to regular TA duties.
- Awarded the Barris Award for Outstanding TA.

Service and Programs

Program Committee / Reviewer, Reliable ML from Unreliable Data, NeurIPS Workshop 2025
Volunteer, 65th IEEE Symposium on Foundations of Computer Science (FOCS) 2024
Ross Mathematics Program (Asia), student, junior counselor, counselor 2018 – 2020
Directed Reading Program, Linear Representations of Finite Groups Spring 2021
Read with Wenyuan Li.

Skills

Technology: C++, Python (NumPy, PyTorch, pandas, matplotlib), Mathematica, and Gurobi Optimizer.